National Institute of Technology Karnataka, Surathkal

Department of Computer Science and Engineering

Interactive Wall Calendar Component

Enhanced Frontend Engineering Challenge Submission

Submission Summary

This report documents an enhanced version of the original frontend engineering challenge. The final project extends the brief into a production-minded interactive calendar experience with polished UX, responsive design, accessibility improvements, reminders, notes, and refined interface behavior.

Submitted By	Aryan Bokolia
Student ID	241CS111
Program	B.Tech. in Computer Science and Engineering
Institution	National Institute of Technology Karnataka, Surathkal
Location	Karnataka, India
Submission Type	Official Frontend Engineering Challenge Report
Project Stack	Next.js, TypeScript, Tailwind CSS, Framer Motion, date-fns
Submission Date	April 8, 2026

1 Project Overview

This project began as a frontend engineering challenge to build an interactive wall calendar component. The implementation was deliberately extended beyond the base requirements to demonstrate stronger frontend engineering judgment across interaction design, accessibility, responsiveness, visual polish, and real-world usability.

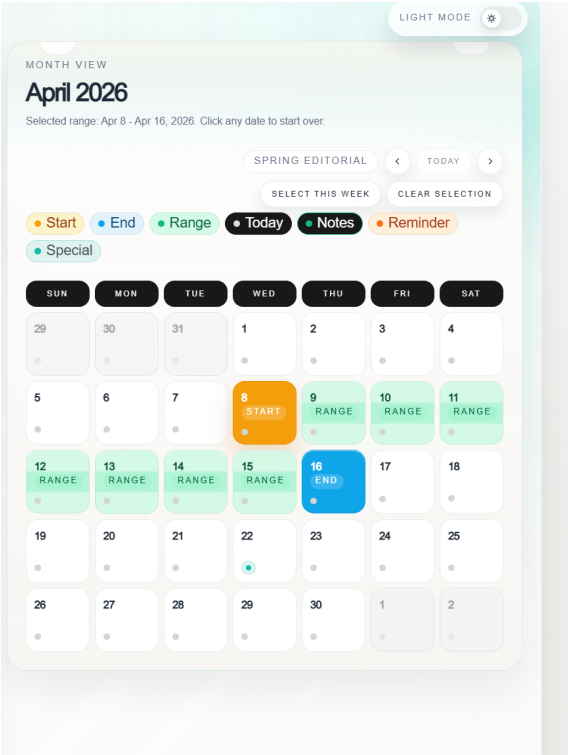
The final product combines a wall-calendar-inspired hero layout, an interactive month grid, a persistent notes workflow, reminder scheduling, motion design, theme support, and robust edge-case handling. The goal was not only to complete the assignment, but to produce a submission that feels product-ready and professionally presented.

2 Submission Context and Objectives

- Build a visually distinctive wall-calendar interface rather than a generic calendar widget.
- Implement date range selection with intuitive interaction patterns, including drag selection.
- Add practical product features such as notes, reminders, notifications, and quick actions.
- Ensure responsive behavior across desktop and mobile form factors.
- Improve accessibility through keyboard navigation, visible focus states, and readable contrast.
- Present the project as an official, recruiter-ready engineering submission.

3 Interface Showcase

(a) Hero section with wall-calendar styling and editorial framing.



(b) Month view with selected range, quick actions, and date markers.

Figure 1: Primary calendar experience.

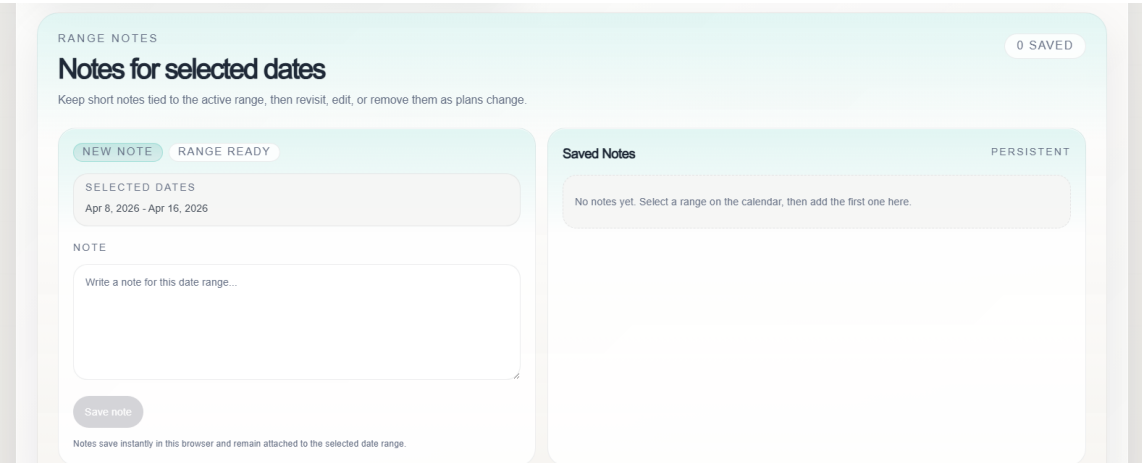


Figure 2: Notes workflow tied to the currently selected date range.

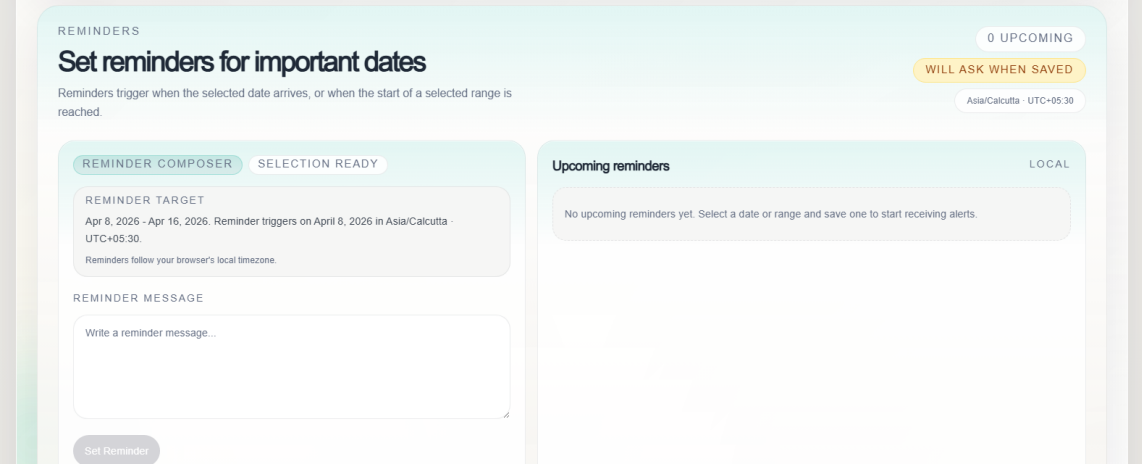


Figure 3: Reminders workflow with timezone-aware targeting and notification status.

4 Implemented Features

4.1 Core Features

- Wall calendar layout with a hero image and structured month-view interface.
- Current month display using a seven-column calendar grid.
- Highlighting for the current day and selected range.
- Fully responsive layout for desktop, tablet, and mobile devices.

4.2 Interaction and Productivity Features

- Click-based date range selection.
- Drag-to-select interaction inspired by booking and scheduling platforms.
- Quick actions including Today, Clear Selection, Select This Week, and month navigation.
- Notes system with create, edit, delete, and localStorage persistence.
- Reminders with browser notifications and in-app fallback alerts.
- Timezone-aware reminder labeling and scheduling behavior.

4.3 Visual and UX Enhancements

- Framer Motion animations for load states, date selection, and month transitions.
- Light and dark mode support with persisted user preference.
- Dynamic accent styling influenced by the hero image.
- Important-date indicators for today, notes, reminders, and special dates.
- Toast-style feedback for note and reminder actions.

4.4 Accessibility and Edge-Case Handling

- Keyboard navigation using arrow keys, Enter, and Space.
- Visible focus states and improved screen-reader clarity.

- Validation for invalid selections, empty note content, and incomplete reminder input.
- Graceful behavior for reversed ranges, same-day selection, and denied notification permissions.

5 Design and Engineering Decisions

5.1 Why the wall calendar concept was chosen

The wall calendar format gives the project a stronger identity than a standard date picker. It creates a more memorable submission and makes the interface feel like a designed product rather than a default utility component.

5.2 Why localStorage was used

The challenge was frontend-focused, so localStorage was selected to keep the application self-contained, easy to run, and simple for reviewers to evaluate without needing backend infrastructure, authentication, or cloud services.

5.3 How UX was approached

The interface was refined with an emphasis on smooth interaction flow, reduced friction, clear feedback, and realistic usage patterns. This includes responsive spacing, drag selection, sticky or full-width workflow sections, action feedback, mobile-friendly controls, and restrained motion.

6 Technology Stack

- Next.js 16 with App Router
- TypeScript
- Tailwind CSS
- Framer Motion
- date-fns
- localStorage for client-side persistence

7 Future Improvements

- Backend integration for cloud persistence and cross-device synchronization.
- User authentication and personalized calendar state.
- Cloud-based reminders with more reliable background delivery.
- Shared planning, collaboration, and multi-user calendar workflows.

8 Conclusion

This submission represents an enhanced and more production-minded interpretation of the original frontend engineering challenge. The project demonstrates not only visual implementation skill, but also thoughtful UX decisions, accessibility improvements, maintainable architecture, and practical product-oriented feature design.

Overall, the work reflects a strong frontend engineering focus with attention to usability, interaction quality, and professional presentation.